



AMMAN CHOUGLE 'S PORTFOLIO

INTRODUCTION

Amman Chougale

Born on 2nd April 2001

Email: amaan020401@gmail.com

Contact Number: +44 7341 328706

Country of origin: India

Open for UK and global opportunities



SUMMARY

Mechatronics Engineer with 3 years of experience transforming ideas into field-ready automation, embedded, and IoT solutions. I specialize in developing practical control architectures, connected hardware, reliable commissioning, and data-driven systems that enhance operational visibility, efficiency, and performance in real-world environments. I am proficient in managing projects effectively to ensure timely delivery while maintaining high quality standards. I also prioritize health and safety by applying structured risk assessment practices to support safe, efficient, and reliable operations.



EDUCATIO

N

MASTER'S DEGREE

MSc (Engineering) in Mechatronics and Robotics,

University of Leeds,

Leeds, United Kingdom

Passed with Merit

September 2022 to December 2023

Modules: Aerial Robotics, Control Systems Design, Soft and Continuum Robotics, Modern Industry Practise, Medical Electronics and E-health, Mechatronics Design and Application, Bio mechatronics and Medical Robotics, Basic Programming



UNIVERSITY OF LEEDS

BACHELOR'S DEGREE

Bachelors in Mechatronics Engineering,

Terna Engineering College(Mumbai University), India

Passed with Distinction

August 2019 to June 2022

Modules: Product development, Sustainable Engineering Practices, Fluid Mechanics, Sensors and Actuators, Modelling and Simulation, Applied Hydraulics & Pneumatics, Renewable Energy Systems, Environmental Engineering Ethics & Policy, Digital Library

DIPLOMA

Diploma in Mechanical Engineering,

Abdul Razzak Kalsekar Polytechnic(MSBTE), India

June 2016 to June 2019

Modules: Mechanical Design, Automotive Design, Heat Ventilation and Air Conditioning, Renewable Energy Systems, Engineering Mechanics and Drawing, Applied Hydraulics and Pneumatics, Basic and Applied Electronics



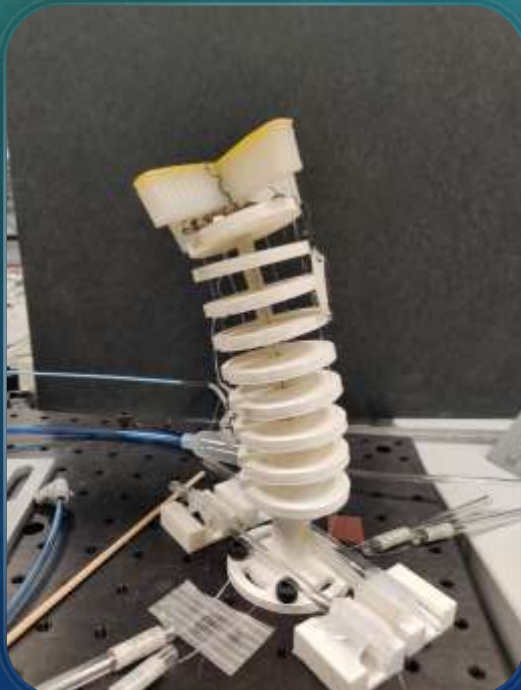
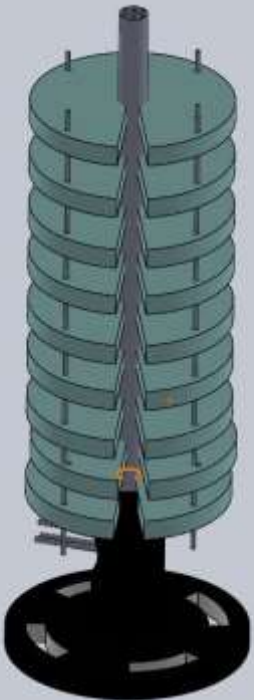
ANJUMAN - I - ISLAM'S
A. R. KALSEKAR
POLYTECHNIC, NEW PANVEL



**PROJECTS
ACHIEVED**

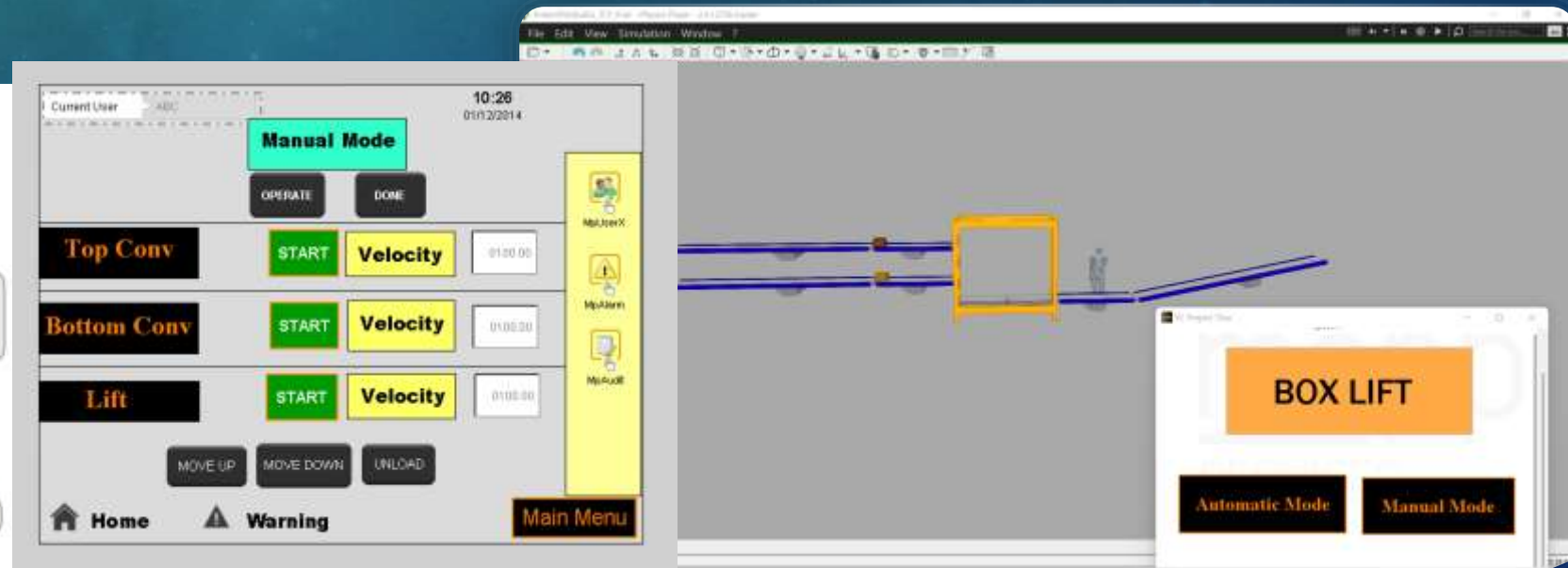
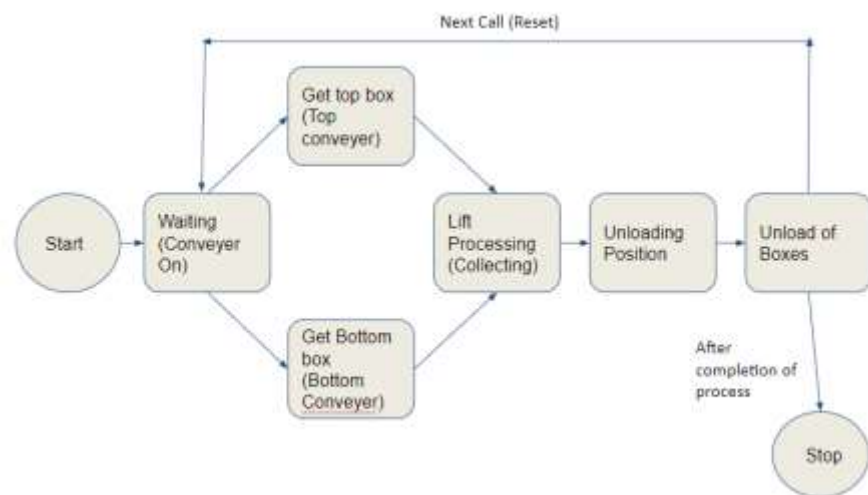
Design and development of continuum arm for soft applications

- ❑ Designed a modular hybrid continuum arm with tendon-driven and pneumatic actuation.
- ❑ Utilized 3D printing and innovative pneumatic tubing for streamlined functionality.



Design And Development Of HMI For Conveyor Systems

- ❑ Developed an automated box lift system with state machine programming, sensor integration, and intuitive HMI.
- ❑ Used digital twin tech for efficient testing, boosting throughput and enabling proactive maintenance.



All terrain trike' (off-road vehicle) during diploma (group of 8)

- ❑ Developed a hybrid crawler-wheeled trike offering enhanced stability, traction, and shock absorption for hostile terrains.
- ❑ Designed for versatile applications in agriculture, construction, and off-road exploration, using cost-effective, existing materials.



TECHNICAL SKILLS

Mechanical Design & CAD – SolidWorks, AutoCAD, Autodesk Inventor, Fusion 360, DFM/DFA, 3D Printing

Hardware Development – Embedded Systems, PCB Design, Circuit Design, Prototyping, Arduino, Raspberry Pi, ESP32/ESP8266

Industrial Automation – PLC Programming (IEC 61131-3), SCADA/HMI, Control Panels, Instrumentation, Commissioning

Industrial IoT – IIoT Architecture, MQTT, HTTP, Edge-to-Cloud Connectivity, Data Acquisition, Remote Monitoring

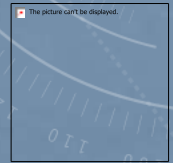
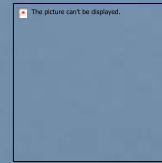
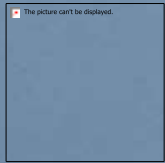
Programming – Python, MATLAB, C, Structured Text (ST), HTML/CSS

Industrial Communication – Modbus RTU/TCP, RS-485, UART, I²C, SPI, Analog & Digital I/O

Engineering & Analysis – System Integration, Testing & Validation, Root Cause Analysis, Feasibility Studies, Technical Documentation

Power & Sustainability – Energy Monitoring, Power Systems, Utility Automation, Net Zero Engineering

INTERPERSONAL SKILLS



Presentation Excellence:
Deliver compelling reports and presentations.

Analytical Insight: Skillfully solve problems and dissect complex issues.

Investigative Strength:
Possess a knack for in-depth research.

Machine Proficiency:
Expertly operate a variety of machinery

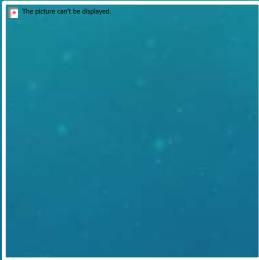
Health/Safety and Risk Assessment:
Implementation

Training:
Software Applications, Embedded, IOT and Mechatronic Systems

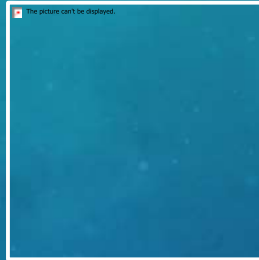


PROFESSIONAL EXPERIENCE

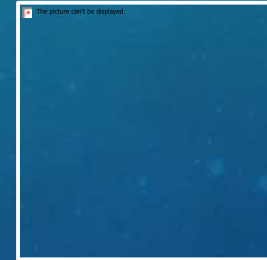
Mechatronics Engineer , Indra Water (Dec 2024- Present)



Industrial IoT, Edge-to-
Cloud Integration &
Automation Lead



Embedded Hardware
Development & Electronic
System Design



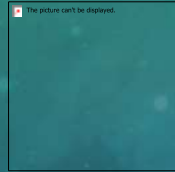
System Integration,
Commissioning &
Troubleshooting

INDRA

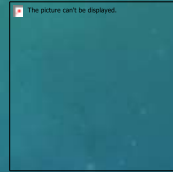
Trainee Project Engineer , B&R Automation Pvt Ltd, Mumbai (Sep 2021 – May 2022)



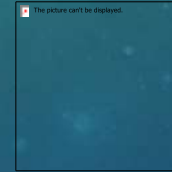
Programming
of B&R control
systems and
software tools.



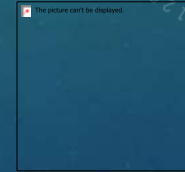
ST and LD
programming.



Digital Twin

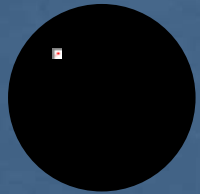


HMI



Automation

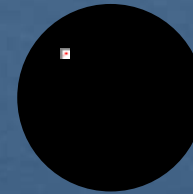
Trainee, Festo Didactic India Pvt. Ltd, Mumbai (July 2020 – Oct 2020)



Hydraulic and
Pneumatic Circuit
Designs



Industrial automation
and robotics on 3D
virtual software.



Basics of PLC and
PLC programming

Trainee Engineer, L&T Electrical And Automation Pvt Ltd, Mumbai (Dec 2019 – Jan 2020)



SR/CT Defect
Analysis



Quality
Improvement



SAP
Optimization



ACB
Assembly/Trouble-
shooting



Team
Collaboration

Schneider
Electric

Lauritz
Knudsen
Electrical & Automation

Additional Experiences

- **Retail Assistant** at Chera Food Company, Leeds, United Kingdom (Nov 2023 – June 2024)
- **Retail Representative** at Marks and Spencer Foods, Leeds, United Kingdom (Mar 2023 – Sep 2023)
- **Mathematics Tutor** at Edufarm Tutorials, Mumbai, India (Apr 2021 – Sep 2022)
- **Document Controller** at I.K. Enterprises (Jan 2020 – Present)

Interests

Mechatronics Engineer

Hardware Design /
Electronics Engineer

Embedded / IOT
Engineer

Medical Devices and
Innovation

Travelling

Cricket

Reading

CONCLUSION

- Give my 100% anywhere I work

- Love working with a team and can work individually

- Always thrive to deliver projects on time



THANK YOU